Fall 2024: CSE 435 **Formal Languages and Automata** **Akdeniz University**

November 15, 2024 Midterm Duration: 90 minutes

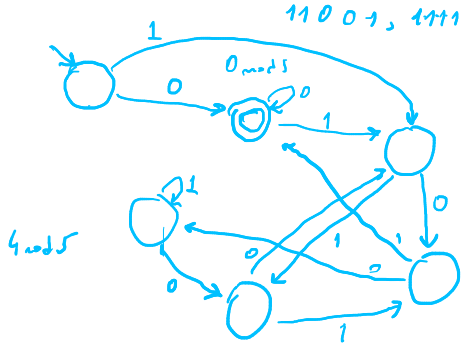
Name: Student No:

P1 **[20 points]** Construct a DFA for the languages

(a) $\{w \mid w \in \{0,1\}^*$ and w has even number of 0s and contains at most two 1s. $\}$

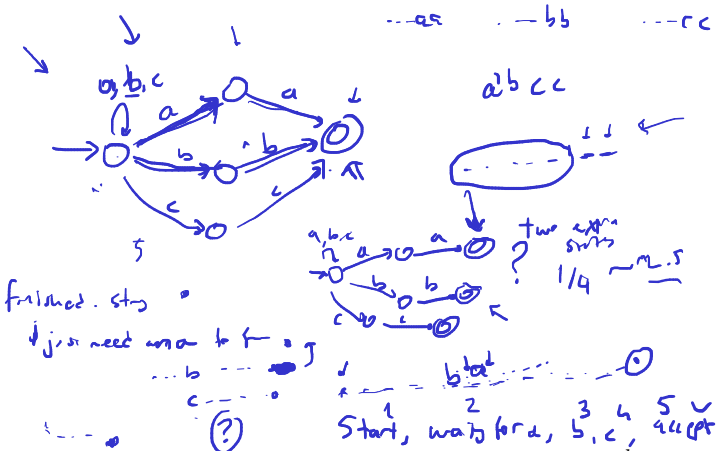


(b) $\{w \mid w \in \{0,1\}^*$ where w is a binary number divisible by 5. $\}$

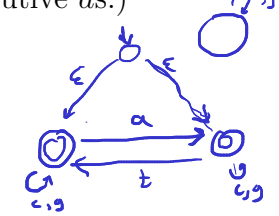


P2 **[20 points]** Construct an NFA for the languages below. Use as few states as you can. Fewest gets full score, every extra state halves your score.

(a) $L = \{w \mid w \in \{a,b,c\}^*, |w| >= 2$, and the last two symbols of w are the same.$\}$



(b) $L = \{w \mid w \in \{a,c,g,t\}^*$ where the symbols a and t alternate when the symbols c and g are ignored $\}$. For example $catgatgga, cggc, \varepsilon \in L$ but $cgatgacaat \notin L$ (because $ataat$ has two consecutive as.)



P3 **[15 points]** Prove that the language $\{0^k 1^m 0^n \mid$ where $k > m > n > 0$. $\}$ is not regular.

Proof. Assume that L is regular. So, by PL for RL, $\exists p \in \mathbb{Z}^+$ s.t. $\forall s \in L$ with $|s| \geq p$, we can split s into 3 pieces as $s = xyz$ s.t. the following conditions are all met: 1. $\forall i \geq 0$ $xy^iz \in L$ 2. $|y| > 0$ 3. $|xy| \leq p$

Now let $s = \underbrace{0^{p+2}}_{x}\underbrace{1^{p+1}}_{y}\underbrace{0^p}_{z}$. To satisfy cond 2 and 3, $y = 0^k$ $k \geq 1$ lying in the first 0 sequence. But then we pump down, $s' = 0^{p+2-k}1^{p+1}0^p \notin L$ since $p+2-k \leq p+1$ (since $1 \leq k$). So, L is not reg. QED.

P4 [30 points] Give the CFG and PDA for the following language: $\{a^k b^m c^n \mid k, m, n \geq 0, k \geq m+n\}$

CFG(15pts):

$S \to aSc \mid B$

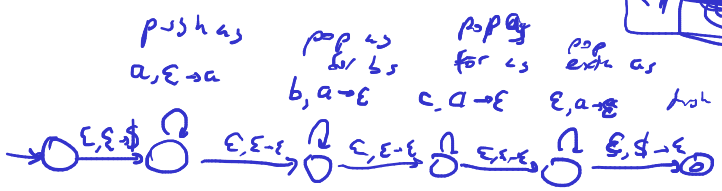
$B \to aBb \mid A$

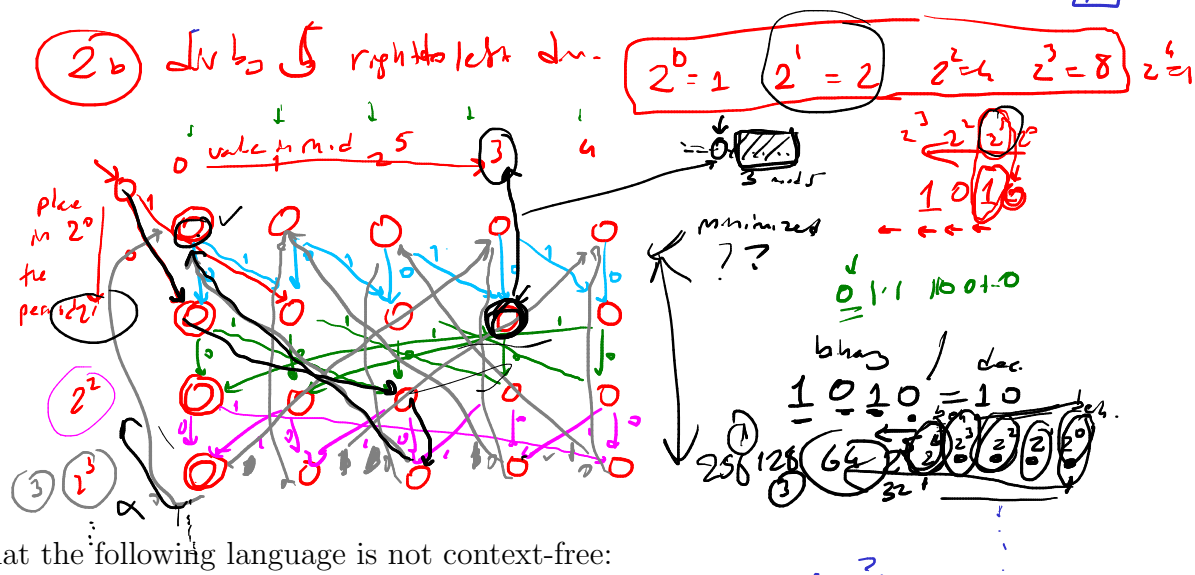
$A \to aA \mid \varepsilon$

$S \to aaa\,B\,ccc$

$S \to B \to A \to \varepsilon$

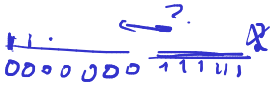
PDA(15pts):





P5 [15 points] Prove that the following language is not context-free:

$$L = \{0^m 1^n 0^k \mid k, m, n \geq 0, k = mn\}$$



Proof. Assume that L is CF. Then, by PL for CFL, $\exists p \in \mathbb{Z}^+$ s.t. $\forall s \in L$ with $|s| \geq p$, s can be split into 5 pieces as $s = uvxyz$ s.t. all following 3 conditions are met.

1. $\forall i \geq 0$ $uv^i xy^i z \in L$ 2. $|vy| > 0$ 3. $|vxy| \leq p$

Now, let $s = 0^p 1^p 0^{p^2}$ ($\in L$, $\geq p$). Indeed. Note that to keep multiplication correct (to get p^2)



1) forever, we need to select v from 1 seq. and y from the second 0 seq. and it must be the case that $p \cdot |v| = |y|$ but due to cond 2, $|v|$ and $|y|$ cannot be zero at the same time. Therefore the minimum possible lengths are $|v| = 1$ $|y| = p$ but then cond 3 is no longer met. Indeed. $|vxy| \geq p+1$.

Thus we're done. L is not C.F. QED.

!!! After you selected s, you must guarantee that however the split is done the string will eventually go out of the lang. after certain number of pump up ops or pumping